

A comparison of the effect of carbon dioxide and relative humidity sensitive ventilation systems on indoor air quality, hygrothermal and energy performance in an office room made of hemp concrete

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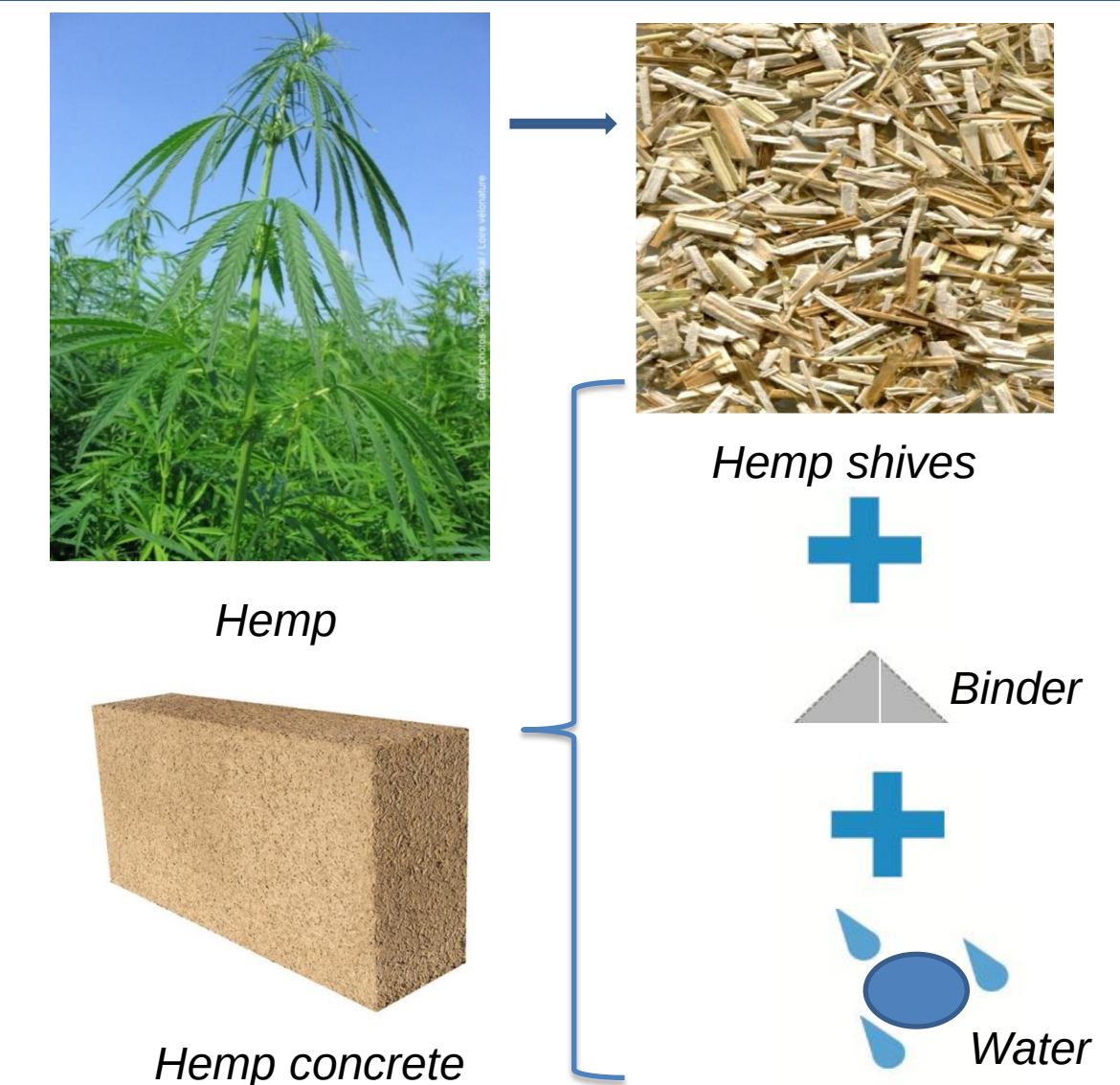
Introduction

❖ Ventilation is one of the most important factors for assuring hygrothermal comfort, good indoor air quality (IAQ) and safety in a building. A too low ventilation rate results in poor IAQ for the occupants and moisture risk, however, too high ventilation rate results in unnecessary energy consumption

❖ Ventilation designed to the occupied zone can be reasonably ascertained by ensuring that the indoor CO₂ concentration is 1000 ppm or less and indoor relative humidity is maintained in the range of 30 to 60 % between winter and summer

❖ Hygroscopic material can regulate the indoor relative humidity variation thanks to its moisture buffering capacity

❖ This article focuses on the effect of carbon dioxide and relative humidity sensitive ventilation systems on indoor air quality, hygrothermal and energy performance in an office room made of hemp concrete



Physical models and numerical solution

Governing moisture balance equation and boundary conditions

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(D_r \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial x} \left(D_\theta \frac{\partial \theta}{\partial x} \right)$$

$$-\rho_l \left(D_r \frac{\partial T}{\partial x} + D_\theta \frac{\partial \theta}{\partial x} \right)_{x=0,e} = h_{M,e} (\rho_{ve,a,e} - \rho_{ve,s,e})$$

$$-\rho_l \left(D_r \frac{\partial T}{\partial x} + D_\theta \frac{\partial \theta}{\partial x} \right)_{x=L,i} = h_{M,i} (\rho_{ve,s,i} - \rho_{ve,a,i})$$

Energy balance equation and boundary conditions

$$\rho_0 C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(\lambda_{app} \frac{\partial T}{\partial x} \right) + L_v \rho_l \left(\frac{\partial}{\partial x} \left(D_{r,v} \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial x} \left(D_{\theta,v} \frac{\partial \theta}{\partial x} \right) \right)$$

$$-\lambda_{app} \frac{\partial T}{\partial x} - L_v \rho_l \left(D_{r,v} \frac{\partial T}{\partial x} + D_{\theta,v} \frac{\partial \theta}{\partial x} \right)_{x=0,e} = h_{T,e} (T_{a,e} - T_{s,e}) + L_v h_{M,e} (\rho_{ve,a,e} - \rho_{ve,s,e}) + \Phi_{ray,e}$$

$$-\lambda_{app} \frac{\partial T}{\partial x} - L_v \rho_l \left(D_{r,v} \frac{\partial T}{\partial x} + D_{\theta,v} \frac{\partial \theta}{\partial x} \right)_{x=L,i} = h_{T,i} (T_{s,i} - T_{a,i}) + L_v h_{M,i} (\rho_{ve,s,i} - \rho_{ve,a,i}) - \Phi_{ray,i}$$

Energy and mass balance equations for room air

$$\rho_i C_p V \frac{\partial T}{\partial t} = \Phi_{West} - \Phi_{East} + \Phi_{South} - \Phi_{North} + \Phi_{Bottom} - \Phi_{Top} + \Phi_{Source}$$

$$V \frac{\partial \rho_i}{\partial t} = Q_{mWest} - Q_{mEast} + Q_{mSouth} - Q_{mNorth} + Q_{mBottom} - Q_{mTop} + Q_{mSource}$$

Model for prediction of indoor carbon dioxide

$$V \frac{\partial C}{\partial t} = Q[C_o(t) - C(t)] + G(t)$$

Prediction of percentage perceived discomfort (PD)

$$PD = \frac{\exp[-0.18 - 5.28(-0.0333H + 1.662)]}{1 + \exp[-0.18 - 5.28(-0.0333H + 1.662)]}$$

Numerical solution

❖ Previous equation system solved using the finite difference technique with an implicit scheme

❖ Discretized equation implemented in the environmental simulation SPARK (Simulation Problem Analysis and Research Kernel)

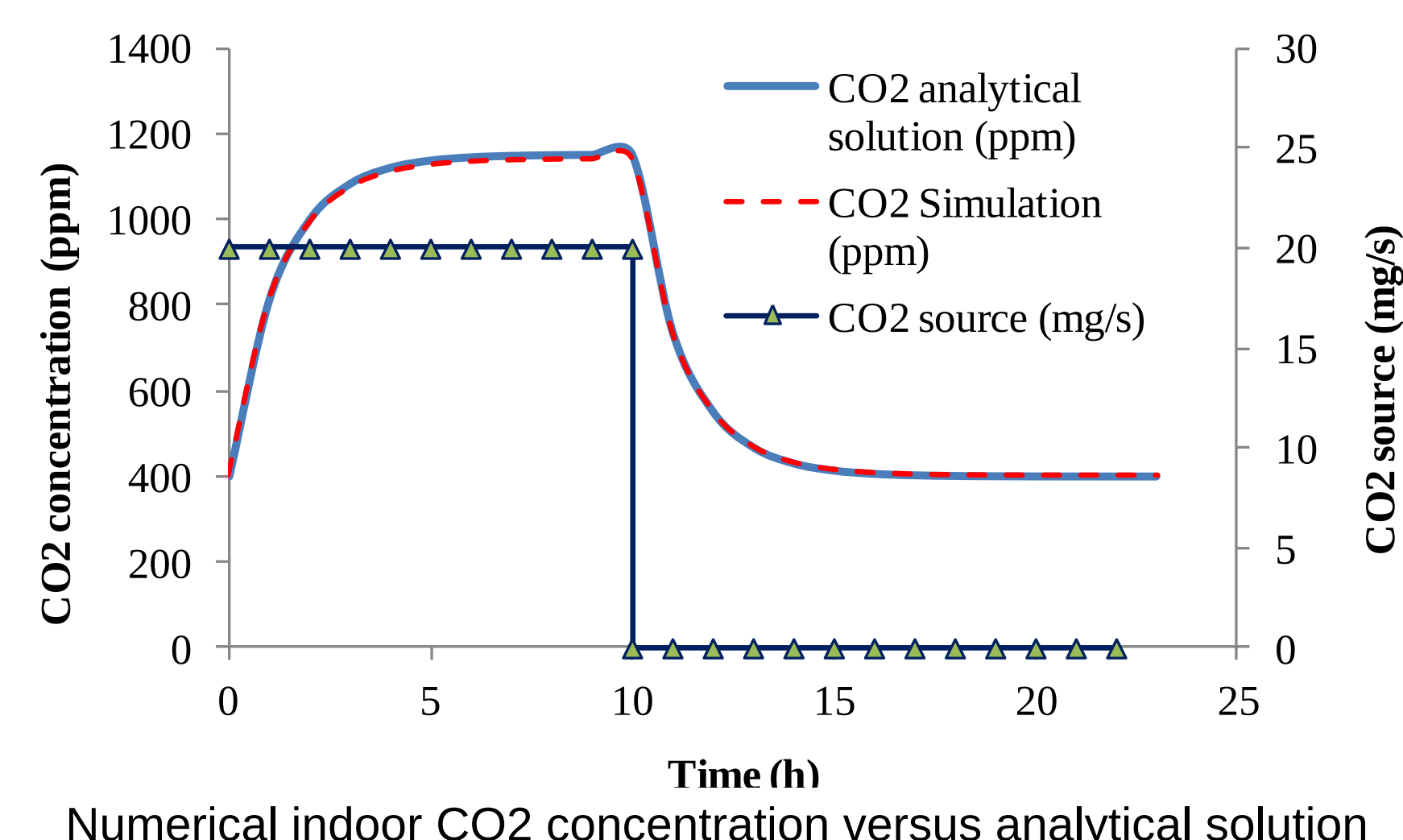
Model validation

Analytical solution

$$C(t) = K.e^{-\frac{Q}{V}t} + \frac{QC_o + G}{Q}$$

Simulation conditions

➤ A room with a volume of 62 m³
➤ Occupied by 2 persons (internal CO₂ source is 20 mg/s) during 10 hours.
➤ Ventilation rate and outdoor CO₂ concentration are 0.8 ach and 400 ppm, respectively.

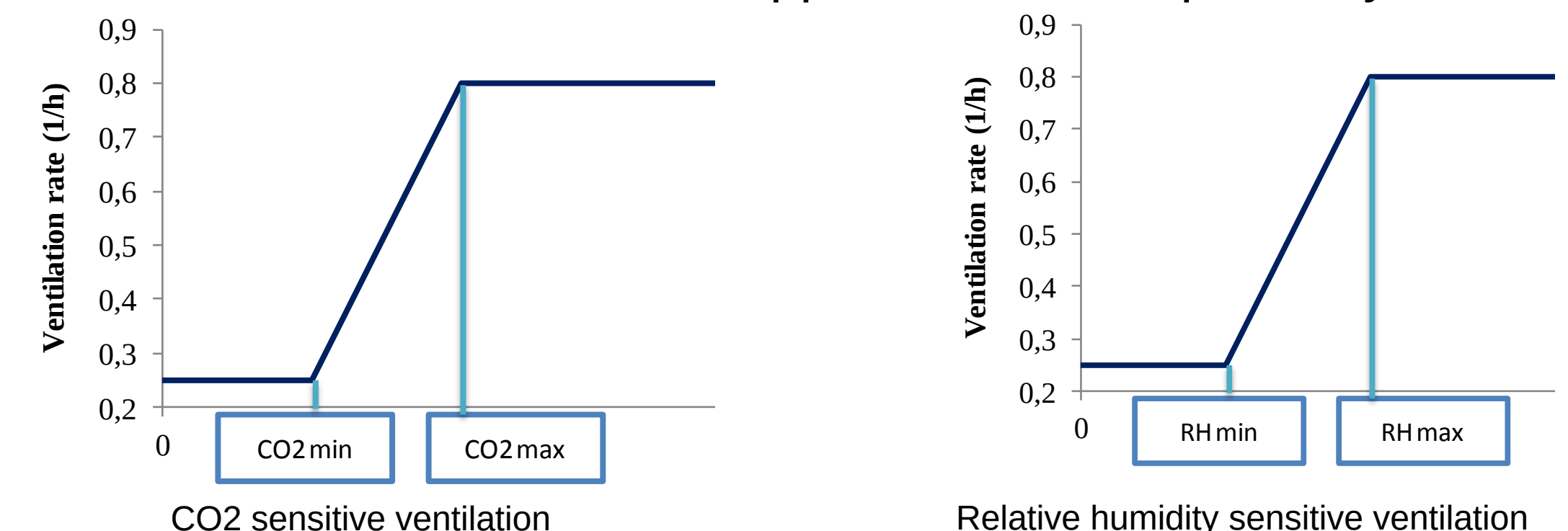


Numerical indoor CO₂ concentration versus analytical solution

Sensitive ventilation system

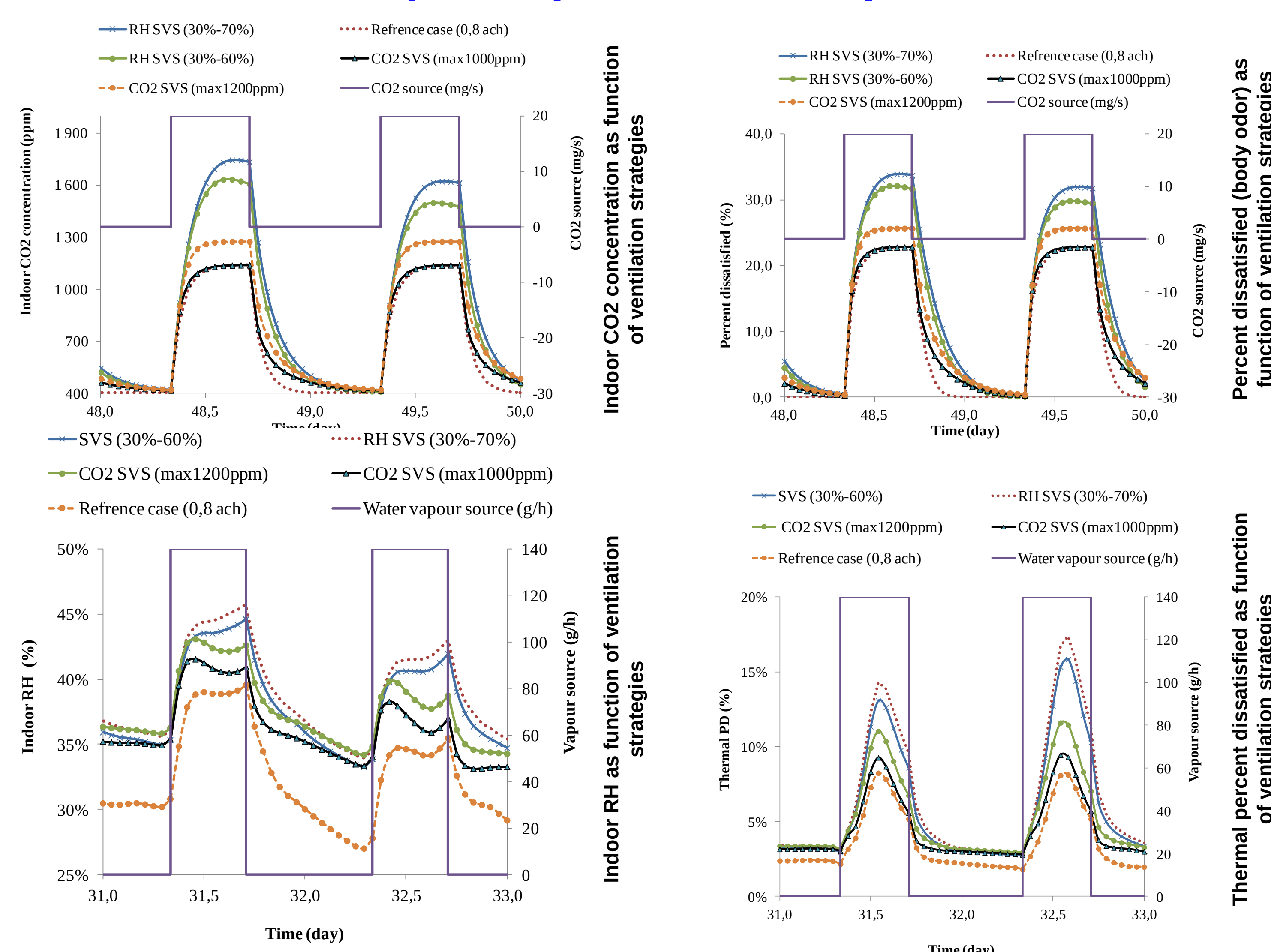
➤ Occupied period: indoor CO₂ or RH level lower than or equal to a maximum target value (CO₂max or RHmax).

➤ Unoccupied period: ventilation rate is maintained at 0.25 ach when the indoor CO₂ or RH lower than 500 ppm or 30 %, respectively.

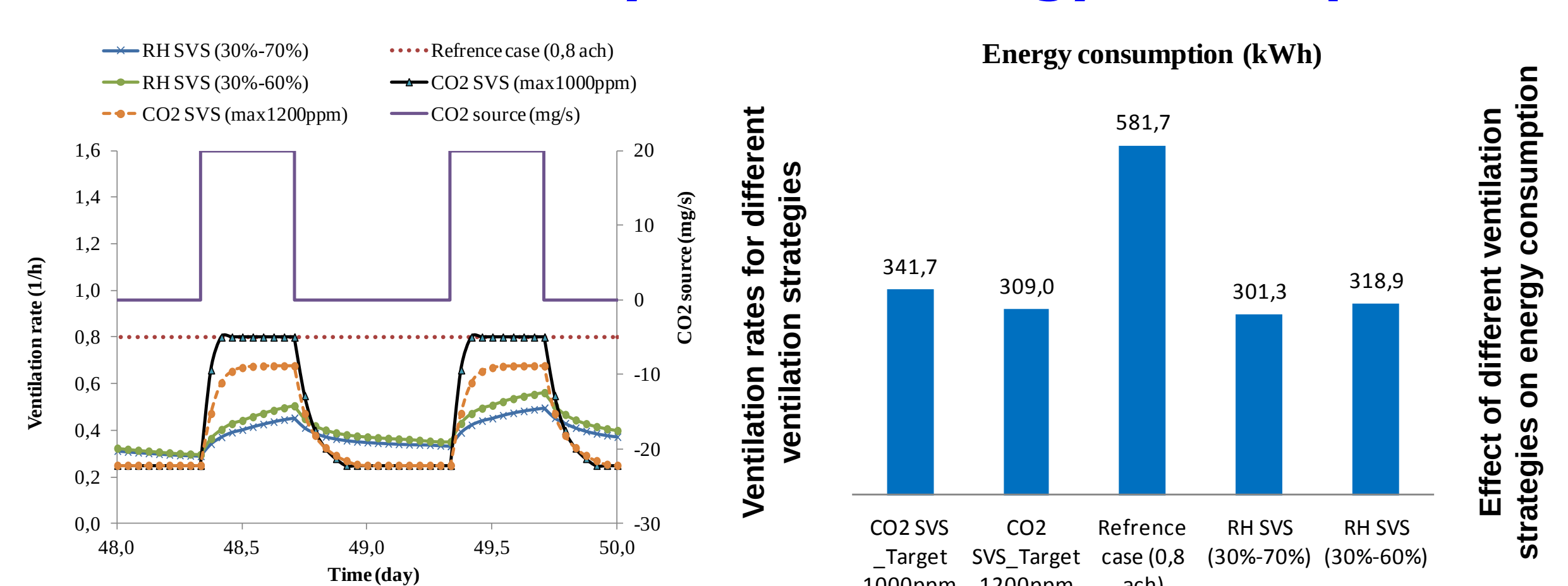


Results

Effect of ventilation systems (CO₂ and RH SVS) on indoor CO₂ and PD



Effect of ventilation systems on energy consumption



Conclusions

❖ A coupled model for predicting hygrothermal performance and indoor air quality in a room is presented and implemented in the environment SPARK

❖ RH SVS widely used in France results in higher percent dissatisfied in terms of body odor and thermal comfort in comparison with the CO₂ SVS because moisture buffering capacity of building materials can reduce indoor RH variations

❖ Use of demand control ventilation system is a very efficient way to reduce the energy consumption (compared to fixed ventilation rate, a reduction of about 48 % without degrading indoor air quality)